DIM3 IP / Due Diligence

The context

Dim3 is positioned in the Medtech sector where we often speak of medical devices.

Because of their design, medical devices are often at the crossroads of several technical fields such as biology, mechanics, computer science and electronics, and thus require complementary technical skills.

In the race for medical innovation, the protection afforded by intellectual property rights is of paramount importance. However, the highly digitalized Medtech field has an IP position derived from the accepted or unaccepted patentability of its components.

In complement, there are regulations that prevent the patentability of certain medtech products.

Exclusions from patentability in Medtech

Medical procedures
As a reminder, patents on medical procedures are prohibited in more than 80 countries. This prohibition was implemented during the Uruguay Round as part of the planned GATT negotiations on trade-related aspects of intellectual property rights (TRIPS).

Diagnostic methods -
Some connected health innovations are intended to perform a diagnostic. Under European law, diagnostic methods applied to the human or animal body are excluded from patentability. Methods that have all the following characteristics are excluded from patentability: an investigation phase, a comparison of these data with normal values, the finding of a significant deviation and a deductive decision phase.
The exclusion does not apply to substances and devices for the implementation of these methods.

Therefore, claims to medical devices, computer programs or recording media are potentially patentable.

Therefore, in E-health, the device itself, such as a wearable that measures blood flow and uses the data to diagnose a cardiovascular problem, can be protected. It is more difficult to patent the software and method included in the device, but it is possible if the claim focuses on how the data is collected, analyzed and processed, without reference to the diagnosis.

Computer programs and mathematical methods.
Under European law, computer programs, mathematical methods, plans, principles and methods for performing intellectual activities as such are excluded from patentability. For both the INPI and the EPO, artificial intelligence is a mathematical method implemented by computer, insofar as it is based on computational models and algorithms "used for classification, partitioning, regression and dimensionality reduction".

In some situations, computer programs and mathematical methods are excluded from patentability only if they are claimed "as such". A computer program is protectable if it produces an additional technical effect beyond the normal physical interaction between the program and the computer.

Above all, a distinction must be made between "computer programs" and "computer-implemented methods" which involve technical means (the computer) and therefore have a technical effect. Similarly, a claim to a mathematical method involving technical means or to a device is not excluded from patentability.

POSITION IP DIM3

With the exception of SCORSO, DIM3's other products have disruptive functions. The disruption is essentially ensured by DIM3's ability to aggregate several technologies into a product designed to meet specific needs pathology by pathology. At the current level performance of products do not include potential of help to decision or automated prescription using Artificial intelligence using metadata grabbed by our system.

Being a Start up with an already analyzed patentability of our products , we have been focused on Freddom to operate protection.

Given DIM3's experience in patenting clinical solutions, we used preliminary reports that defined DIM3's overall IP strategy.

DIM3 consulted three patentability-related structures:
- A Freedom to Operate analysis was performed with PiCarré;
- A general patentability analysis was performed with Pécher de Groote;
- A strategic analysis was carried out with the firm Koan & Partners.

Based on this information and its verification by experience or meeting with other specialists, the patentability strategy is as follows.

The deliverable is essentially a platform integrating software, algorithms and medical devices.

Software
Software is difficult to patent and is more a matter of copyright than of patentability. Patentability is only possible at a later stage when the software integrating artificial intelligence will be an autonomous decision support system.

Computer algorithm
A computer algorithm used in a program can be patented. However, we do not patent algorithms. When filing a patent, the algorithm must be included in some detail. It is therefore preferable to have a secrecy strategy rather than a patent strategy

Informatic clinical algorithm
A clinical algorithm could be discovered by analyzing the large amount of data collected by our products. It will be part of a strategic review from the moment Dim3 launches its artificial intelligence analysis project to assist in clinical decision making. Any protection action is premature

Sensor / Data transmitter
DIM3's Feedim is likely to include a number of technical and usability innovations. However, the sensor technology and some applications are already in the public domain. We preferred to have a current silence strategy. The presence of sensors in Feedim could be an asset to try patentability. We did not work on this due to financial reason and lack of experiences.

In conclusion,

e-Depots were filed in 2015 to ensure at a minimum the Freedom to Operate of the Dim3 platform and its components. We have not renewed these e-filings given the presence of our products on the market.

Clinical discoveries from the analysis of connected meta-data as well as Feedim's functions related to software embedded in medical devices will be the subject of strategic repository analysis with Baxter.

Dim3 is particularly attentive to the evolution of the patentability of solutions including artificial intelligence, as the arrival of this technology in force opens many debates on patentability and regulation?